Shivam Kumar

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Summary

SDE II with 5+ years at Microsoft building large-scale cloud-native platforms, distributed systems, and Kubernetes-based container infrastructure on Azure. Deep expertise across microservices architecture, AI/LLM-powered SRE automation and Dapr, a CNCF Graduated project. Recognized for high-impact delivery: IDC Pinnacle FY'26 Best AI Success Story; 2× RockStar Award.

Experience

Software Development Engineer II — Jul 2024 – Present
Microsoft – Azure Container Apps — *Hyderabad, India*

- Engineered AI-powered SRE agent for Azure Container Apps, reducing P50 TTM by **97%** and TTR by **64%**; implemented RCA capabilities, automated self-repair, intelligent resolution, and Ev2 deployment pipelines.
- Enhanced Azure Functions on ACA with graceful draining, secure function key management, KEDA scaling-rule customization, JWT clock-skew tolerance, invocation tracing, and durable-event logging; deployed to ∼**1,322** external apps.
- Drove ACA Express Public Preview on Azure Dev Compute (ADC), resolving P0 blockers via ephemeral volume support and Log Analytics customer-log delivery; unblocked migration path for ∼**100K+** ACA environments.
- Owned ACA Sessions management APIs and resolved critical async execution bugs, improving platform stability for ∼**300** external customers; enabled early session deletion for the Fabric team, optimizing resource utilization and costs.
- Led Dapr AKS Extension rollouts (v1.15, v1.16), remediating CVEs across ∼**400** AKS clusters and integrating CodeQL scanning for enhanced security posture.
- Architected Foundry Control Plane (FCP) integration for ACA agentic apps, supporting both customer-declared and auto-detected agent models; enables ∼**2,000+** agentic applications to be centrally managed via FCP.

Software Development Engineer I — Jul 2021 – Jul 2024
Microsoft – Dapr — *Hyderabad, India*

- Stabilized Dapr building blocks by upgrading Configuration API, MQTT PubSub, CosmosDB state store, and Redis/PostgreSQL Configuration stores; added bulk-subscribe for Azure Event Hubs; enabled Dapr CLI airgap installation.
- Implemented Cascade Terminate/Purge for Dapr Workflows; identified and fixed scalability bottlenecks in Dapr workflows through comprehensive performance testing.
- Co-launched ACA Sessions with end-to-end ownership of Node.js and .NET code-interpreter support (adopted by Security Copilot); resolved critical session bugs and added regression tests protecting **3000+** active sessionPools.
- Designed Dapr's internal fork infrastructure across **5** repositories with a patch-based model; built OneBranch-compliant pipelines using Ev2 and automated OSS-to-internal releases with <**1** day SLA.
- Led multi-architecture support for Dapr and dashboard images; migrated to 1P Container Registry, established ACA nightly builds, and added Mariner and distroless images to MCR.

Quantitative Research Intern — May 2020 – Jul 2020
JP Morgan & Chase – E-Trading — *Mumbai, India (Remote)*

- Built price prediction models using **LSTM**, feedforward neural networks, and regression with univariate feature selection; improved predictive power by **40%** for HK names through feature standardization.
- Implemented L1 and L2 features for optimizing model parameterization across multiple asset classes.

Research

Data Race Detection for Task-Parallel Programs — Jun 2019 – Aug 2021
Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed novel algorithm **Tasker** integrating vector clocks with tree-based techniques; implemented via **LLVM** memory instrumentation pass on task-parallel programs; achieved **1.48× speedup** over PTRacer with lower memory overhead on 128GB Intel Xeon.
- *"Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks"* — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

Projects

GemOS | *C, x86 — OS kernel (IIT Kanpur CS330)* — Aug 2019 – Nov 2019

- Implemented mmap/munmap/mprotect with lazy allocation and page-fault handling; cfork/vfork with copy-on-write semantics.

Machine Learning | *Python, Keras, ML (CS771)* — Aug 2019 – Nov 2019

- CNN image classification using Keras; SGD for binary classification; multi-label Bonsai recommender system.

Cipher Decoder | *Python, Modern Cryptology (CS641)* — Aug 2020 – Nov 2020

- Differential cryptanalysis of 3-DES; decryption of Caesar, Vigenere ciphers using frequency analysis.

Education

Indian Institute of Technology, Kanpur — CGPA: 9.4/10.0
Bachelor of Technology in Computer Science and Engineering — *2017 – 2021*

Technical Skills

Languages: Go, C#/.NET, Python, C/C++
Cloud & Infra: Azure(ACA, Functions, AKS), Kubernetes, Docker, Helm, KEDA
Distributed Systems & Platforms: Dapr (CNCF), Microservices, Service Mesh, gRPC, Pub/Sub, Event-Driven Architecture

Achievements & Recognition

IDC Pinnacle FY'26 — Best AI Success Story (ACA RCA Agent squad, ∼200+ competing teams)
RockStar Award, Microsoft ('22, '23) for exceptional engineering contributions
Academic Excellence Award (×2), IIT Kanpur | JEE Advanced **AIR 715** | JEE Mains **AIR 348** | KVPY Scholar | NSE Physics **Top 1%**